

Research Statement

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My research focuses on provable and practical quantum advantages in communication and streaming models. I combine quantum algorithm design with computational complexity techniques to understand how quantum computing offers substantial advantages over classical computing.

Three motivations guide my research.

1. Prominent quantum algorithms, including Shor’s algorithm [Sho97] and decoded quantum interferometry (DQI) [JSW⁺25], have primarily been studied in time complexity. Space and communication are also fundamental computational resources. Space efficiency is particularly important given the substantial overhead of implementing fault-tolerant logical qubits [FMMC12].
2. Rapid advances in quantum hardware motivate the search for problems amenable to experiments on NISQ devices or practical demonstrations of quantum advantage. Quantum communication and streaming offer promising directions for realizing advantages with modest quantum resources. For example, Kumar et al. demonstrated a photonic implementation of Sampling Matching that uses less transmitted information than the best-known classical protocol [KKD19], and Kretschmer et al. demonstrated an unconditional advantage in information resources using only 12 trapped-ion qubits, performing a task that provably requires at least 62 bits of classical memory [KGD⁺25].
3. Shor’s algorithm and DQI just offer superpolynomial speedups over the best-known classical algorithms. Communication and streaming models provide settings in which such advantages can be proved unconditionally [GKK⁺08, KPV24].

Guided by these motivations, my doctoral research has developed along three closely related directions:

- The first exponential quantum advantage in multiparty communication [YZ26, WWY26], resolving questions posed by Gavinsky and Pudlák [GP08] and Göös, Gur, Jain, and Li [GGJL25].
- The exponential quantum space advantages for approximating Max- k SAT in the streaming setting and a classification of Boolean Max-2CSPs admitting quantum advantages [WY26a].
- The unconditional quantum space advantages for decoded quantum interferometry in the streaming setting [WY26b], which imply the first exponential one-pass quantum streaming advantages over multipass classical streaming algorithms for natural problems.

Quantum advantages in multiparty communication model

A fundamental objective in the study of quantum advantage is to exhibit computational separations across diverse models. Communication complexity has seen substantial progress toward this goal: a line of work [BCW98, Raz99, BYJK04, GKK⁺08, Gav16, GGJL25] now establishes exponential separations in several two-party communication settings. Despite these successes in two-party communication complexity, quantum advantage in the multiparty setting remains poorly understood. The premier model for multiparty communication is the Number-on-the-Forehead (NOF) model, introduced by Chandra, Furst, and Lipton [CFL83]. In this model, the input is partitioned into k parts, and player i can see all parts except their own (as if they were “written on their forehead”). Recently, Göös, Gur, Jain, and Li [GGJL25] highlighted the following open problem:

Is there an exponential quantum advantage in Number-on-Forehead communication?

Resolving this problem would demonstrate that quantum communication maintains an exponential advantage in a multiparty setting with highly overlapping input views. The main obstacle to obtaining strong separations between quantum and randomized protocols in the NOF model, already noted in [GGJL25], is the lack of techniques for proving lower bounds for randomized NOF protocols. Most existing methods, such as the generalized discrepancy method [RY15], apply equally well to quantum and randomized communication, making it difficult to distinguish the power of quantum versus classical protocols in the NOF setting [LS09].

My Work

As a first step toward understanding quantum advantage in NOF communication, I studied its one-way variant, in which each player speaks exactly once in a fixed order. This is an open problem proposed by Gavinsky and Pudlák [GP08]. In joint works with Jiapeng Zhang (FOCS 2026), I developed a new lifting framework that connects two-party communication to one-way NOF communication. Using this framework, we obtained the first exponential separation between quantum and randomized one-way NOF communication [YZ26], resolving an open problem posed by Gavinsky and Pudlák [GP08].

This line of research culminated in subsequent work with Haoyu Wang and Pei Wu, where we got the first exponential quantum advantage in the multiparty communication model, answering the open problem proposed by [GGJL25].

In related work with Xudong Wu and Penghui Yao (ICALP 2026), I also developed a lifting theorem for hybrid classical–quantum communication complexity, establishing new trade-offs between classical and quantum communication costs [WYY26].

Quantum space advantages in streaming setting

Quantum computing offers the possibility of substantial resource advantages over classical computation. For combinatorial optimization, a long line of work has provided some evidence of possible superpolynomial quantum speedups for certain optimization problems [FGRV25, LZW23, CLZ22, PUW⁺24, Sze22, GHV21, EH22]. However, establishing an unconditional superpolynomial quantum advantage in time complexity for optimization remains extremely challenging.

Space is another central resource, especially when logical qubits carry substantial fault-tolerant implementation overhead [FMMC12]. Streaming algorithms provide a natural framework for studying space-efficient quantum computation and for seeking quantum advantages in space usage [NT17]. Motivated by processing massive datasets, in the streaming model, data elements arrive sequentially, and the algorithm must process each element as it appears while using only limited working space. Because of their space efficiency, streaming algorithms have found many applications in the computation of statistics over data streams and the estimation of parameters of massive graphs.

Early work gives exponential quantum advantages in one-way communication to obtain analogous space advantages for specially constructed streaming problems [GKK⁺08]. More recent work extended this phenomenon to natural optimization problems, showing that Maximum Directed Cut, a natural Boolean Max-2CSP, admits an exponential quantum space advantage [KPV24]. This development raises a broader classification question:

Which Max-CSPs admit exponential quantum space advantages in the streaming model?

My Work

Motivated by this open question, I studied quantum space advantages for Boolean Max-CSPs, starting with Max- k SAT. Max- k SAT is one of the most basic Boolean CSPs, and classical streaming lower bounds of Chou, Golovnev, and Velusamy [CGV20] imply that any algorithm achieving an approximation ratio better than $\sqrt{2}/2 \approx 0.7071$ requires $\Omega(\sqrt{n})$ space.

In joint work with Haoyu Wang, I proved exponential quantum space advantage for Max- k SAT [WY26a]. We give a one-pass quantum streaming algorithm using $\text{polylog}(n)$ space that achieves a 0.7425-approximation on instances with n variables. Combined with the classical lower bound of [CGV20], this gives an exponential separation between quantum and classical streaming space for Max- k SAT. Together with known results, this gives a complete classification of which Boolean Max-2CSPs admit exponential quantum space advantages.

In joint work with Minbo Gao, Chenghua Liu, and Tianyi Zhang (SODA 2027), I developed a new technique for proving quantum streaming lower bounds via matrix discrepancy [GLYZ26]. Our approach gives quantum streaming lower bounds for triangle finding. This direction complements the algorithmic side of quantum streaming.

Quantum advantages for decoded quantum interferometry

Decoded quantum interferometry (DQI), introduced by Jordan et al. [JSW⁺25], connects approximate optimization to classical error correction. For optimal polynomial intersection (OPI), a special case of Max-LINSAT, Reed–Solomon decoding yields worst-case approximation guarantees exceeding those of known polynomial-time classical algorithms in certain parameter regimes. However, establishing an unconditional superpolynomial quantum advantage in time complexity remains wide open. Space is another central resource, especially when logical qubits carry substantial fault-tolerant implementation overhead. This motivates a natural question:

Can DQI yield an unconditional quantum space advantage for optimization?

My work.

In joint work with Kewen Wu, we established an unconditional quantum space advantage for Hermite Optimal Polynomial Intersection (HOPI), a natural extension of OPI that asks for a polynomial satisfying as many constraints on its values and derivatives as possible [WY26b]. We adapted DQI to the streaming model and obtained an efficient one-pass quantum algorithm with a strong approximation guarantee. In contrast, any one-pass classical algorithm achieving a comparable approximation requires polynomially more memory. At comparable memory, a classical algorithm must make polynomially many passes, regardless of its running time. Thus, a single quantum pass suffices where classical algorithms with comparable memory require polynomially many passes.

The classical lower bounds build on gadgetless lifting, a technique I introduced in joint work with Jiapeng Zhang (STOC 2024) [YZ24]. We used it to prove communication lower bounds for collision problems, resolving open problems in proof complexity and cryptography. I subsequently developed this approach with Xinyu Mao and Jiapeng Zhang (ITCS 2025) to obtain improved lower bounds for pointer chasing [MYZ25], and with Mi-Ying Huang, Xinyu Mao, Shuo Wang, and Jiapeng Zhang (CCC 2025) to establish multi-party communication lower bounds with applications to streaming [HMW⁺25]. My other work (TCC

2023) used density-increment arguments to prove lower bounds for cryptographic key-agreement protocols [HMYZ23]. This line of work provides a methodological foundation for my study of provable quantum advantages.

Future Research Directions

Building on these results, I plan to pursue two complementary directions that connect provable quantum advantages with practical implementations.

From provable space advantages to provable quantum speedups

I aim to identify OPI-related optimization problems that admit polynomial-time quantum algorithms based on DQI while remaining hard for classical polynomial-time algorithms under explicit complexity-theoretic or cryptographic assumptions. The central challenge is to connect these assumptions to the algebraic structure exploited by DQI. Beyond analyzing existing algorithms, I also aim to develop new quantum algorithms for natural problems, combining classical hardness analysis with concrete resource estimates to identify real-world settings in which exponential speedups could yield practical benefits.

Toward practical quantum streaming algorithms

I plan to develop implementation-oriented variants of our Max- k SAT algorithm and related Max-DiCut algorithms. A proof-of-principle implementation of the quantum pair sketch for Hidden Matching on a trapped-ion processor provides a starting point [NCK⁺25]. The accompanying fault-tolerant resource estimates highlight circuit depth as a major obstacle to practical advantage. I will focus on reducing logical-qubit requirements and circuit depth, improving noise robustness, and minimizing classical-control latency, seeking to translate asymptotic space savings into feasible implementations.

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